

Tay, L., Meade, A. W., & Cao, M. (in press). An overview and practical guide to IRT measurement equivalence analysis. *Organizational Research Methods*. doi: 10.1177/1094428114553062

Walkthrough for Illustrations

Illustration 1

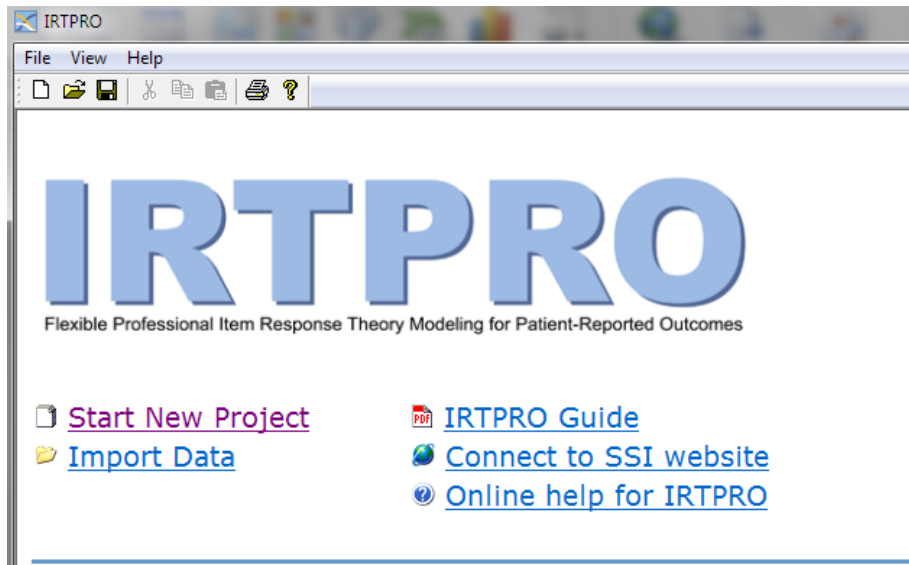
File Name	Comment
Simulated_DData.csv	Contains simulated data of 2000 individuals. Group = 1 represents the reference group (N = 1000); Group = 2 represents the focal group (N = 1000); I1 to I15 represents items 1 to 15. See the simulated item parameters below (Table 8 in paper).
Simulated_DData.irtpro	IRTPRO syntax file
Simulated_DData.SSIG	IRTPRO data file (converted from the .csv file)
Simulated_DData.Model0-irt	Model 0 Output – Simultaneous estimation (no constraints)
Simulated_DData.Model1-irt	Model 1 Output – Fully constrained model
Simulated_DData.Model2-irt	Model 2 Output – Testing anchor items with two-step procedure
Simulated_DData.Model3-irt	Model 3 Output – Testing non-anchor items for DIF
Simulated_DData.Model4-irt	Model 4 Output – Further testing non-anchor items for DIF

Table 8. Illustration 1: Simulated item and theta parameters

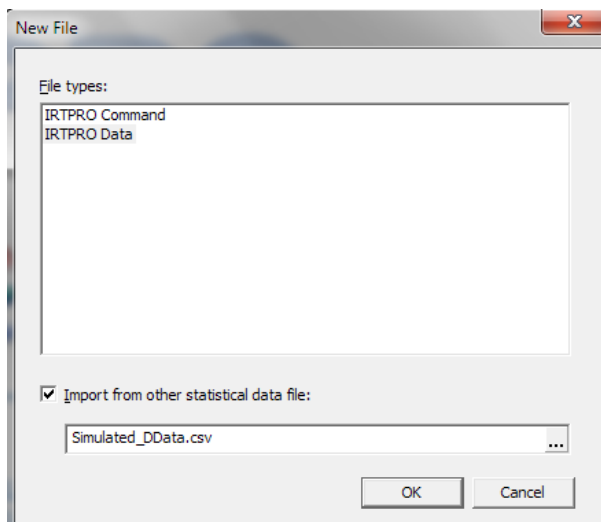
	Group = 1 ($\theta_{\text{mean}}=0, \theta_{\text{SD}}=1$)				Group = 2 ($\theta_{\text{mean}}=0.2, \theta_{\text{SD}}=1$)				<u>Type of DIF</u>
	λ	γ	$\underline{a}$	$\underline{b}$	λ	γ	$\underline{a}$	$\underline{b}$	
1	0.90	-0.26	2.06	-0.29	0.90	-0.26	2.06	-0.29	Large ab DIF
2	0.66	-0.06	0.88	-0.09	0.66	-0.06	0.88	-0.09	
3	0.83	-0.42	1.49	-0.51	0.43	0.08	0.48	0.19	
4	0.71	-0.14	1.01	-0.20	0.71	-0.14	1.01	-0.20	
5	0.77	-0.37	1.21	-0.48	0.77	-0.37	1.21	-0.48	Large ab DIF
6	0.68	-0.34	0.93	-0.50	0.68	-0.34	0.93	-0.50	
7	0.58	-0.48	0.71	-0.83	0.18	0.02	0.18	0.11	
8	0.80	-0.07	1.33	-0.09	0.80	-0.07	1.33	-0.09	
9	0.85	-0.3	1.61	-0.35	0.85	-0.3	1.61	-0.35	Large ab DIF
10	0.85	-0.48	1.61	-0.56	0.85	-0.48	1.61	-0.56	
11	0.82	-0.27	1.43	-0.33	0.42	0.23	0.46	0.55	
12	0.8	-0.26	1.33	-0.33	0.8	-0.26	1.33	-0.33	
13	0.85	-0.03	1.61	-0.04	0.85	-0.03	1.61	-0.04	Large ab DIF
14	0.84	-0.14	1.55	-0.17	0.84	-0.14	1.55	-0.17	
15	0.86	-0.27	1.69	-0.31	0.46	0.23	0.52	0.50	

STEP 1: Creating SSIG file (IRTPRO data file)

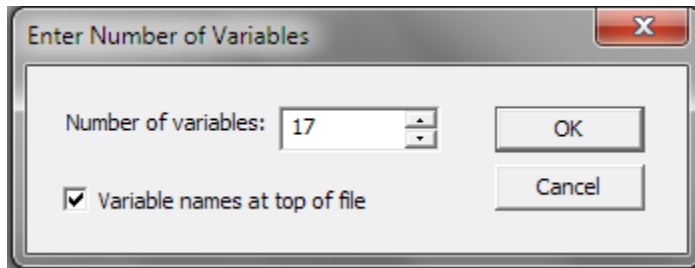
A. Click on “Start New Project”



B. Select the data file. In this case, we have “Simulated_DData.csv” as our raw data. Then click “OK”

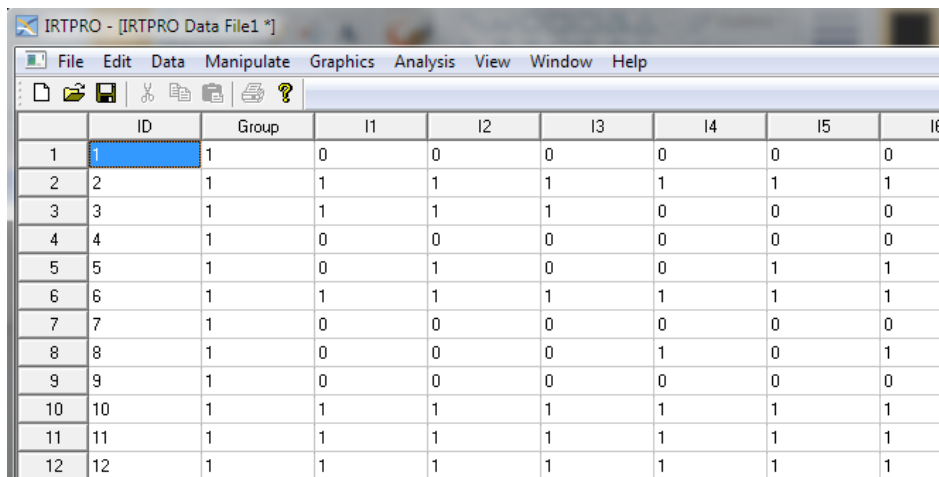


- C. We have 17 variables here: “ID”, “Group”, and 15 item responses. Also we have our “Variable names at the top of the file”. Click “OK”



The dialog box titled "Enter Number of Variables" has a close button (X) in the top right corner. It contains a label "Number of variables:" followed by a text box containing the number "17" and a small up/down arrow button. Below this is a checked checkbox labeled "Variable names at top of file". To the right of the checkbox are two buttons: "OK" and "Cancel".

- D. Check that the data is correctly read in.

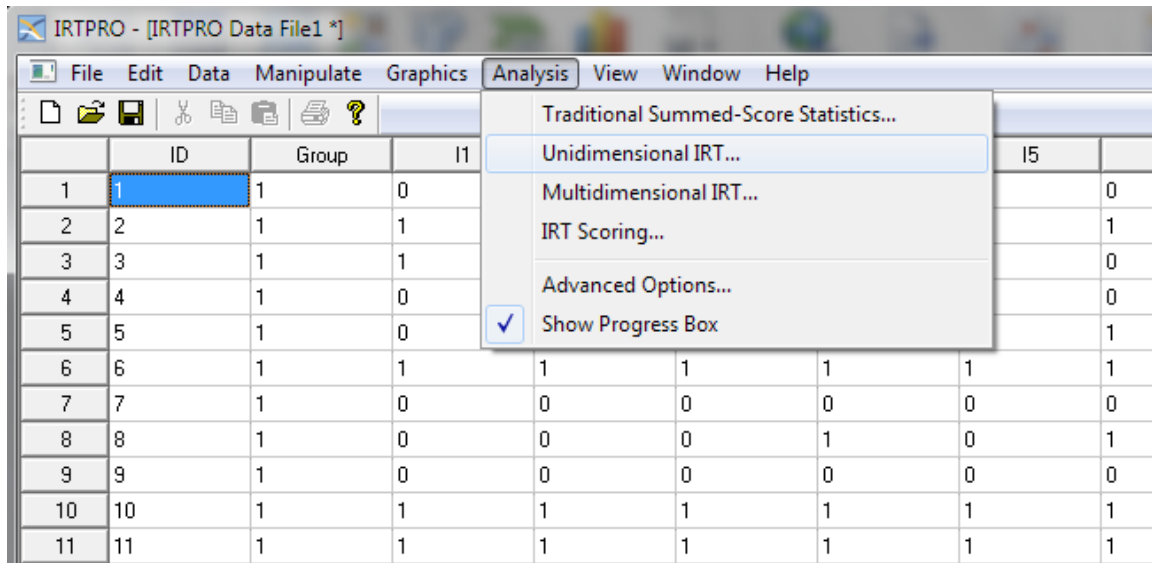


The screenshot shows the IRTPRO software window with the title "IRTPRO - [IRTPRO Data File1 *]". The menu bar includes File, Edit, Data, Manipulate, Graphics, Analysis, View, Window, and Help. Below the menu bar is a toolbar with icons for file operations and a help icon. The main area displays a data table with 9 columns: ID, Group, I1, I2, I3, I4, I5, and I6. The table contains 12 rows of data. The first row (ID 1) is highlighted in blue.

	ID	Group	I1	I2	I3	I4	I5	I6
1	1	1	0	0	0	0	0	0
2	2	1	1	1	1	1	1	1
3	3	1	1	1	1	0	0	0
4	4	1	0	0	0	0	0	0
5	5	1	0	1	0	0	1	1
6	6	1	1	1	1	1	1	1
7	7	1	0	0	0	0	0	0
8	8	1	0	0	0	1	0	1
9	9	1	0	0	0	0	0	0
10	10	1	1	1	1	1	1	1
11	11	1	1	1	1	1	1	1
12	12	1	1	1	1	1	1	1

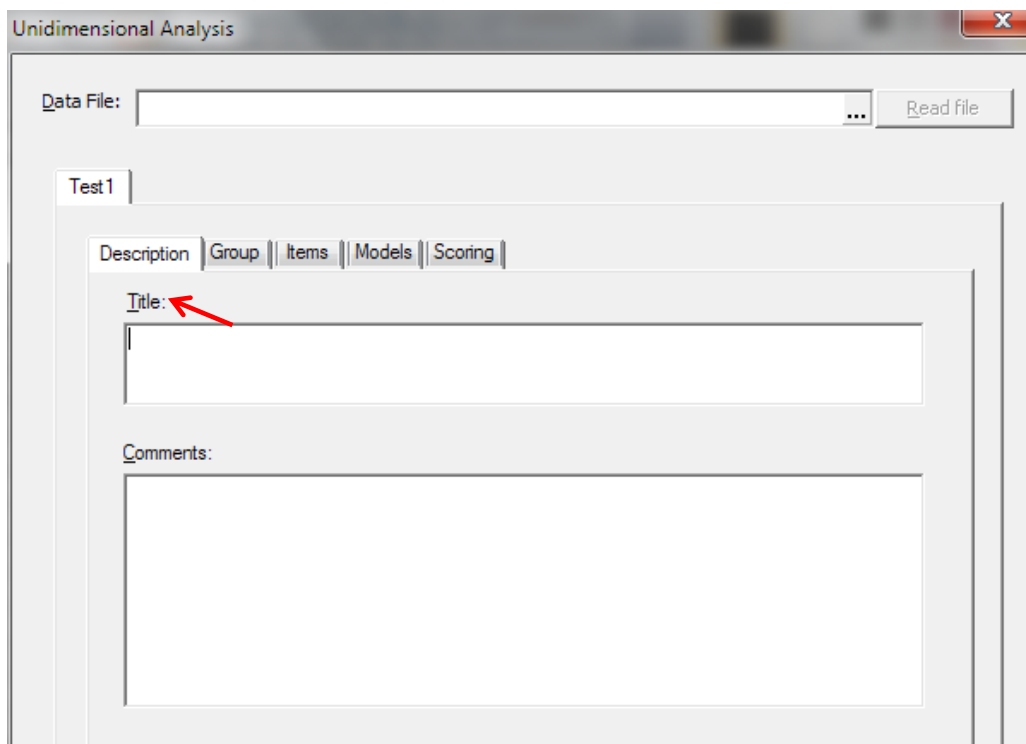
STEP 2: Analyze data using simultaneous estimation (i.e., simultaneous calibration) of both groups (Model 0)

- A. Because we are conducting a unidimensional IRT analysis, we select: Analysis > Unidimensional IRT...

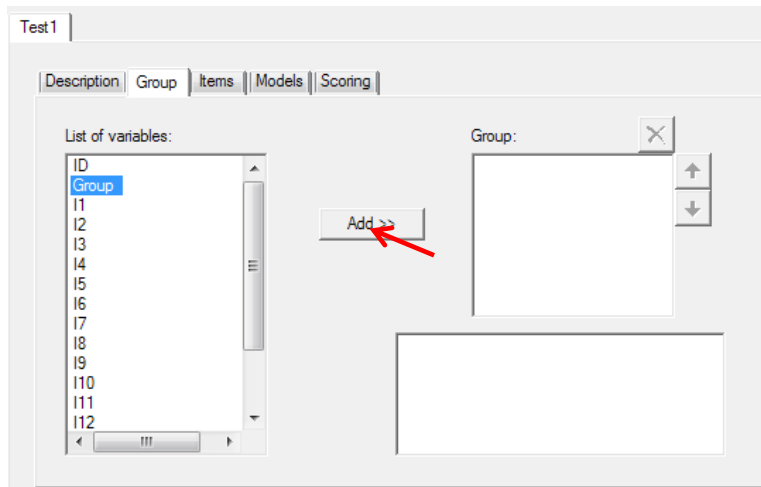


- B. Optional: You can fill in the “Title” for the analysis and “Comments” to keep track of what model you specify.

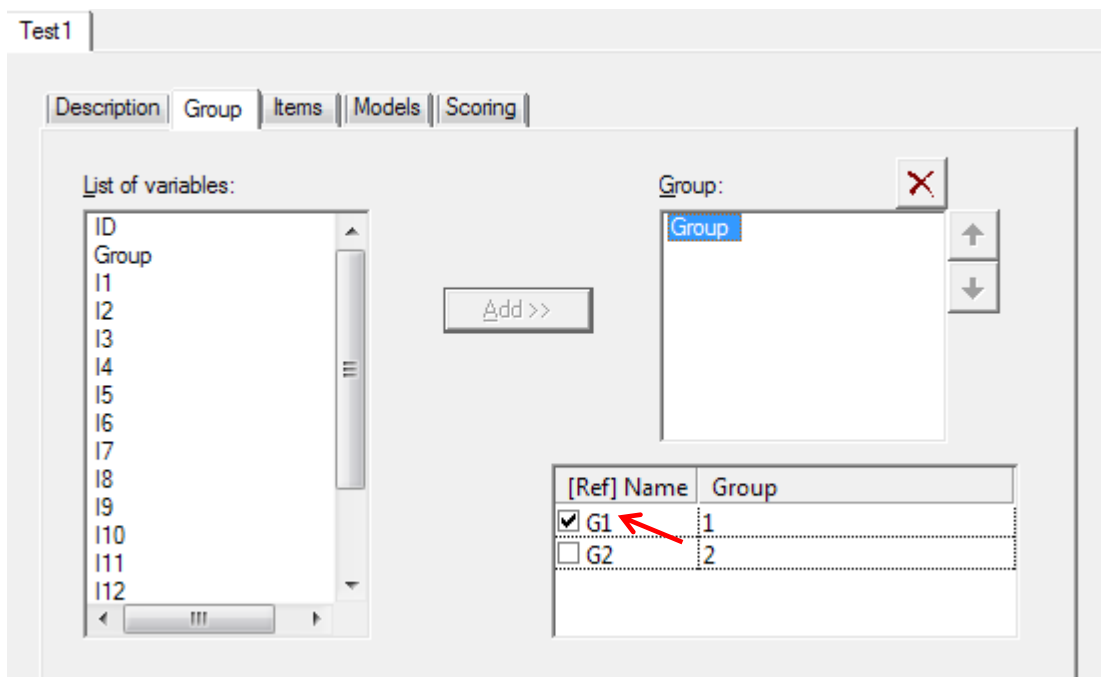
*Note: You do not need to select the data file as that is already selected even though it appears blank



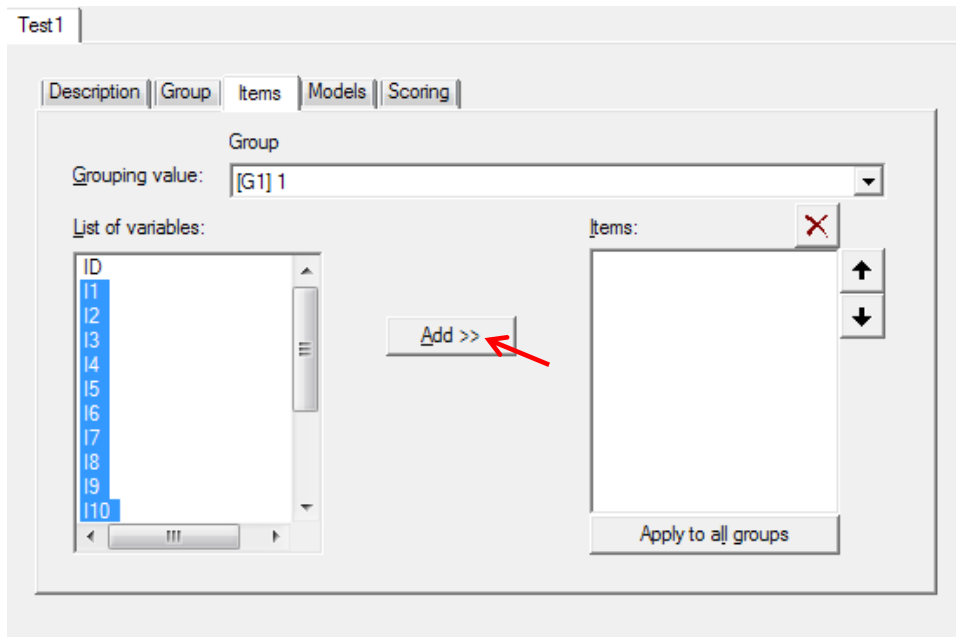
- C. In the “Group” tab, add the “Group” variable to the “Group:” box. This tells IRTPRO that there are multiple groups (≥ 2) in the data.



*Note: The first group is automatically selected as the “reference” group as shown in the “check box”.

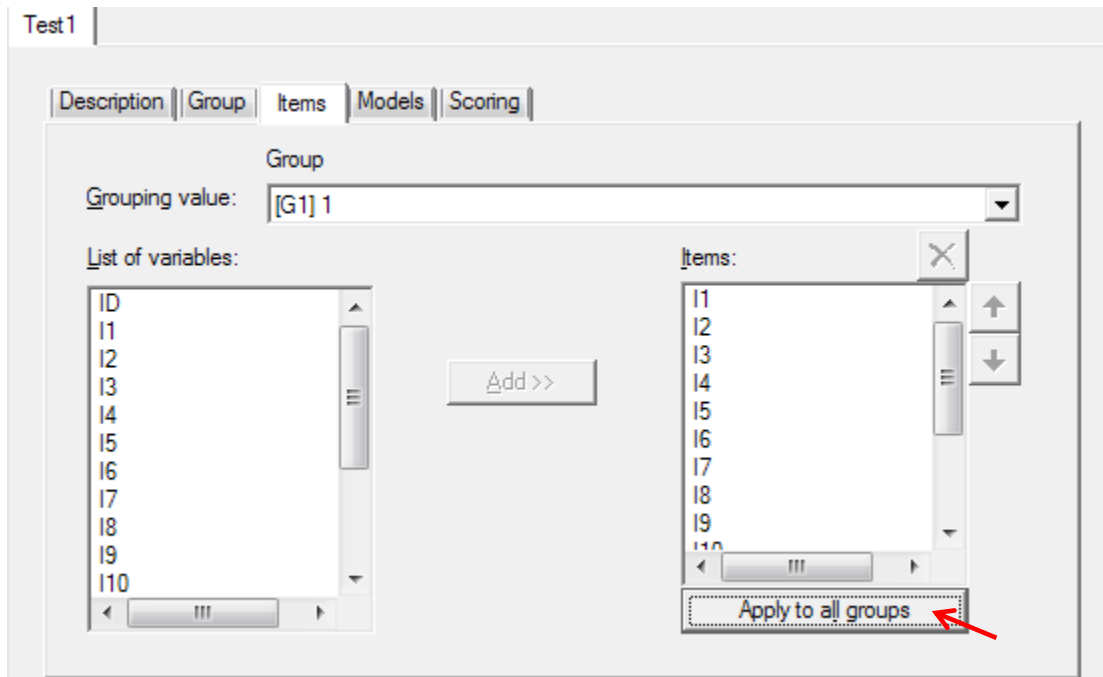


- D. In the “Items” tab, we select all the item variables into the “Items:” box. This tells IRTPRO which items we want to analyze



Then we click on “Apply to all groups”. This tells IRTPRO that the same set of items were administered across both groups (Group 1 and Group 2).

After clicking “Apply to all groups”, a box will appear “Previous settings will be lost. Do you want to continue?”. Click “Yes”



- E. In the “Models” tab, we can specify which items to test for DIF and which items (and item parameters) to constrain as equal across groups. For the first analysis, we do not need to specify and DIF analysis or constraints. We note that because the data are dichotomous the model is “2PL” by default.

Test1

Description | Group | Items | Models | Scoring

Group

Grouping value:

	Item List	Categories	Data Codes	Item Scores	Model
	I1	2	0, 1	0, 1	2PL
	I2	2	0, 1	0, 1	2PL
	I3	2	0, 1	0, 1	2PL
	I4	2	0, 1	0, 1	2PL
	I5	2	0, 1	0, 1	2PL
	I6	2	0, 1	0, 1	2PL
	I7	2	0, 1	0, 1	2PL
	I8	2	0, 1	0, 1	2PL

Constraints... DIF... Apply to all groups

- F. In the “Scoring” tab, we do not need to do anything as we are not interested in scoring participants. However, if one is interested to do so, one should specify the “Person ID”, select the type of scoring method: “EAP” or “MAP scores”. The results of EAP or MAP are quite similar and “EAP” is used more often.

Test1

Description | Group | Items | Models | Scoring

☐ Person ID:

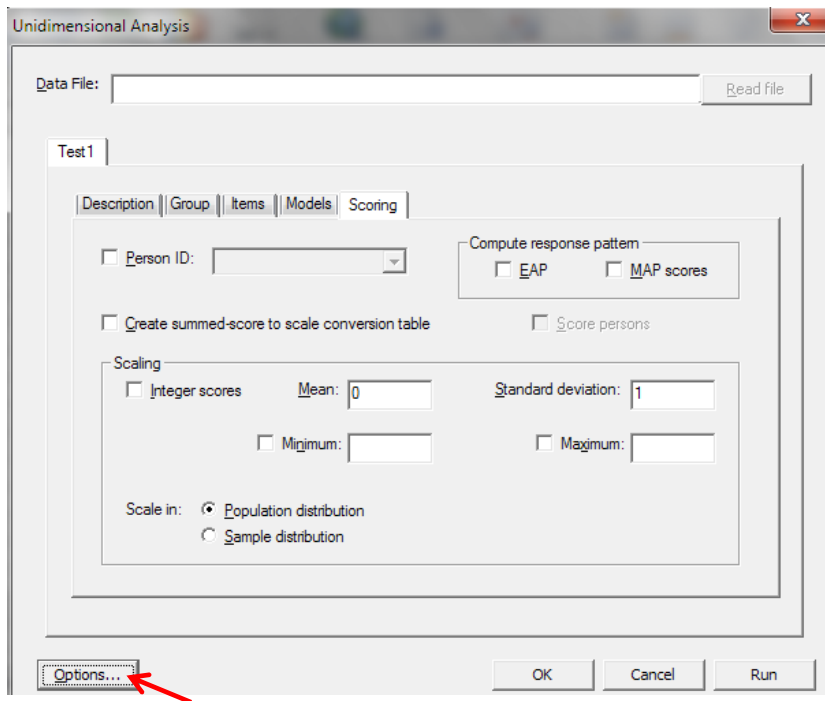
Compute response pattern
☐ EAP ☐ MAP scores

☐ Create summed-score to scale conversion table ☐ Score persons

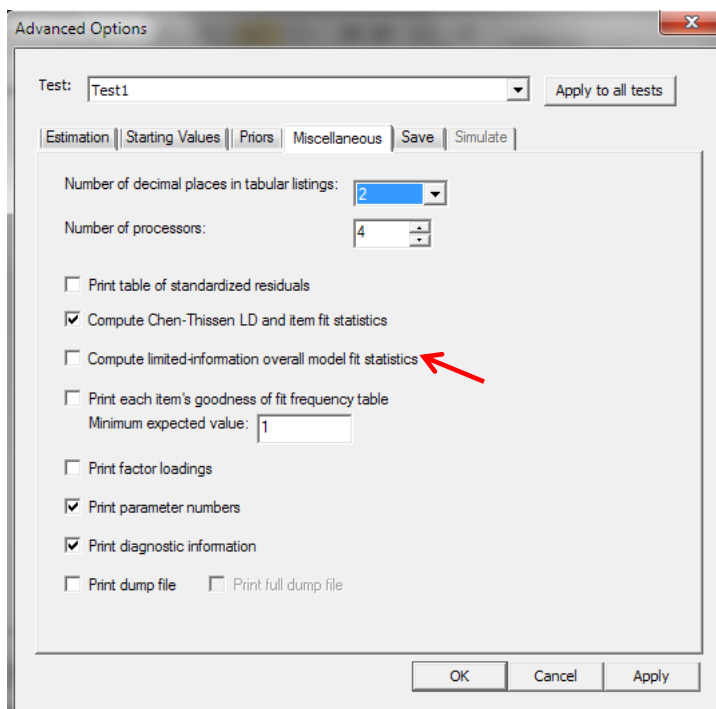
Scaling
☐ Integer scores Mean: Standard deviation:
☐ Minimum: ☐ Maximum:

Scale in: ☒ Population distribution ☐ Sample distribution

G. Finally, to obtain the overall fit statistics (i.e., M_2 and RMSEA), we will need to go into “Options”



In the “Options” menu, select the “Miscellaneous” tab. Check the “Compute limited-information overall model fit statistics”. Note. when checking this box, a text box will appear warning that this can take a long time: “This can take a long time if the number of items and/or dimensions is large.” Click “OK”.



Then “Apply” and “OK”

H. After specifying all the necessary model information we can “Run” the analysis.

Unidimensional Analysis

Data File:

Test1

Description | Group | Items | Models | Scoring

☐ Person ID:

☐ Create summed-score to scale conversion table

Scaling

☐ Integer scores Mean: Standard deviation:

☐ Minimum: ☐ Maximum:

Scale in: ☒ Population distribution ☐ Sample distribution

☐ Compute response pattern

☐ EAP ☐ MAP scores

☐ Score persons

STEP 3: Interpreting output (for Model 0)

The output will be produced in a html format...

IRTPRO Version 2.1

Output generated by IRTPRO estimation engine Version 4.54 (32-bit)

Project:	
Description:	
Date:	28 May 2014
Time:	09:51 PM

Table of Contents

[2PL Model Item Parameter Estimates for Group 1, logit: \$a\theta + c\$ or \$a\(\theta - b\)\$](#)

[Factor Loadings for Group 1](#)

[Summed-Score Based Item Diagnostic Tables and \$\chi^2\$ s for Group 1](#)

[2PL Model Item Parameter Estimates for Group 2, logit: \$a\theta + c\$ or \$a\(\theta - b\)\$](#)

[Factor Loadings for Group 2](#)

[Summed-Score Based Item Diagnostic Tables and \$\chi^2\$ s for Group 2](#)

[Group Parameter Estimates](#)

[Marginal fit \(\$\chi^2\$ \) and Standardized LD \$\chi^2\$ Statistics for Group 1](#)

[Marginal fit \(\$\chi^2\$ \) and Standardized LD \$\chi^2\$ Statistics for Group 2](#)

[Item Information Function Values for Group 1 at 15 Values of \$\theta\$ from -2.8 to 2.8](#)

[Item Information Function Values for Group 2 at 15 Values of \$\theta\$ from -2.8 to 2.8](#)

[Likelihood-based Values and Goodness of Fit Statistics](#)

[Summary of the Data and Control Parameters](#)

Overview:

Content	Comment
-2PL model item parameter estimates for Group 1 -2PL model item parameter estimates for Group 2	2PL item parameter estimates for Groups 1 and 2
-Summed-Score Based Item Diagnostic Tables and χ^2 s for Group 1 -Summed-Score Based Item Diagnostic Tables and χ^2 s for Group 2	This shows the “S- χ^2 ” where we can examine individual item fit
-Marginal fit (χ^2) and standardized LD χ^2 statistics for Group 1 -Marginal fit (χ^2) and standardized LD χ^2 statistics for Group 2	This shows the standardized LD χ^2 statistics we can examine violations of unidimensionality for pairs of items
-Likelihood-based values and goodness of fit statistics	This shows the M_2 and RMSEA value. We can also obtain different information criteria.
-Factor loadings for Group 1 -Factor loadings for Group 2	We specified “Factor Loadings” in the “Options” tab in Step 2G. This produces factor loadings.
-Group parameter estimates	This shows the estimated focal group latent trait distribution (mean & SD). The reference group is usually constrained as $N(0,1)$.
-Item information function values for Group 1 -Item information function values for Group 2	This is the discretized information function for items
-Summary of the data and control parameters	This displays what data were analyzed and estimation information

Some things to note when interpreting the output...

A. The IRTPRO item parameter estimates are:

$$\frac{1}{1 + \exp[-a_i^*(\theta_j - b_i)]}$$

Because we simulated item parameters in with a scaling factor of “1.702”

$$\frac{1}{1 + \exp[-1.702a_i(\theta_j - b_i)]}$$

The IRTPRO item parameter estimates for the a-parameter (a_i^*) is our simulated item parameter a_i multiplied by 1.702

2PL Model Item Parameter Estimates for Group 1, logit: $a\theta + c$ or $a(\theta - b)$ [\(Back to TOC\)](#)

Item	Label	a	s.e.	c	s.e.	b	s.e.
1	I1	2 3.27	0.27	1 1.11	0.18	-0.34	0.05
2	I2	4 1.45	0.12	3 0.22	0.10	-0.15	0.06
3	I3	6 2.67	0.22	5 1.55	0.17	-0.58	0.05
4	I4	8 1.90	0.15	7 0.33	0.11	-0.17	0.06
5	I5	10 2.13	0.17	9 1.14	0.14	-0.54	0.05
6	I6	12 1.74	0.14	11 0.96	0.12	-0.55	0.06
7	I7	14 1.18	0.11	13 1.00	0.09	-0.85	0.09
8	I8	16 2.10	0.16	15 0.41	0.12	-0.19	0.05
9	I9	18 2.78	0.23	17 1.17	0.17	-0.42	0.05
10	I10	20 2.67	0.23	19 1.77	0.18	-0.66	0.05
11	I11	22 2.63	0.22	21 1.16	0.16	-0.44	0.05
12	I12	24 2.25	0.18	23 0.90	0.13	-0.40	0.05
13	I13	26 3.04	0.25	25 0.24	0.15	-0.08	0.05
14	I14	28 2.55	0.20	27 0.63	0.14	-0.25	0.05
15	I15	30 2.43	0.20	29 1.07	0.15	-0.44	0.05

Here, we see that multiplying the simulated a-parameter by 1.702 leads produces values similar to the IRTPRO estimates. We also need to check that the s.e.’s for the items are small showing that the estimates are fairly accurate.

	<u>a</u>	<u>b</u>	<u>a*</u>
1	2.06	-0.29	3.51
2	0.88	-0.09	1.50
3	1.49	-0.51	2.54
4	1.01	-0.20	1.72
5	1.21	-0.48	2.06
6	0.93	-0.50	1.58
7	0.71	-0.83	1.21
8	1.33	-0.09	2.26
9	1.61	-0.35	2.74
10	1.61	-0.56	2.74
11	1.43	-0.33	2.43
12	1.33	-0.33	2.26
13	1.61	-0.04	2.74
14	1.55	-0.17	2.64
15	1.69	-0.31	2.88

- B. The S- χ^2 statistic shows the fit of each individual item. We hope to see that the modeled and the observed frequencies are not significantly different, implying that there is good/reasonable model-data fit. There may be several items that show misfit but a majority of the items should fit well for the specified IRT model. Otherwise, a different model should be considered.

Summed-Score Based Item Diagnostic Tables and χ^2 s for Group 1 [\(Back to TOC\)](#)

S- χ^2 Item Level Diagnostic Statistics

Item	Label	χ^2	d.f.	Probability
1	I1	9.88	12	0.6279
2	I2	8.45	13	0.8136
3	I3	8.98	13	0.7751
4	I4	9.81	13	0.7100
5	I5	21.41	13	0.0651
6	I6	10.11	13	0.6860
7	I7	7.22	13	0.8908
8	I8	7.63	13	0.8672
9	I9	10.88	13	0.6222
10	I10	15.56	13	0.2729
11	I11	10.84	13	0.6251
12	I12	16.30	13	0.2328
13	I13	6.78	12	0.8721
14	I14	9.09	13	0.7671
15	I15	11.88	13	0.5386

- C. Group parameter estimates show the estimated latent trait mean and variance (and sd) for the reference group. In this case G1 is the reference group which has the mean and sd fixed at 0 and 1, respectively.

Group Parameter Estimates [\(Back to TOC\)](#)

Group	Label	μ	s.e.	σ^2	s.e.	σ	s.e.
1	G1	0.00	----	1.00	----	1.00	----
2	G2	0.42	0.27	1.60	0.48	1.26	0.19

D. The standardized LD χ^2 statistics we can examine violations of unidimensionality for pairs of items. Generally, absolute values smaller than 3 indicate good fit. IRTPRO differentiates the magnitude of the standardized LD χ^2 using different shades of colors. Red represents negative associations beyond the single latent trait; blue represents positive associations beyond the single latent trait. Brighter colors indicate larger magnitudes.

Marginal fit (χ^2) and Standardized LD χ^2 Statistics for Group 1 [\(Back to TOC\)](#)

Item	Label	Marginal χ^2	1	2	3	4	5	6	7	8	9	10
1	I1	0.0										
2	I2	0.0	-0.7									
3	I3	0.0	0.3	-0.7								
4	I4	0.0	-0.7	-0.7	-0.6							
5	I5	0.0	-0.7	-0.7	-0.0	-0.7						
6	I6	0.0	-0.6	-0.7	-0.7	0.8	-0.5					
7	I7	0.0	0.2	-0.6	-0.6	-0.7	-0.7	-0.7				
8	I8	0.0	-0.7	-0.7	0.2	-0.6	-0.5	0.3	-0.7			
9	I9	0.0	-0.5	-0.6	-0.7	-0.4	-0.7	-0.4	-0.6	-0.7		
10	I10	0.0	-0.3	-0.7	-0.6	0.4	-0.7	-0.6	-0.6	-0.6	-0.7	
11	I11	0.0	-0.7	-0.6	-0.6	-0.5	-0.7	-0.7	-0.6	-0.7	-0.6	-0.7
12	I12	0.0	-0.6	-0.1	-0.7	-0.5	0.0	-0.7	-0.7	-0.7	-0.5	0.3
13	I13	0.0	0.7	-0.7	-0.6	-0.7	-0.5	0.3	-0.5	-0.7	-0.6	-0.7
14	I14	0.0	-0.7	-0.4	-0.5	0.1	-0.5	-0.5	-0.4	-0.3	1.0	-0.7
15	I15	0.0	-0.6	-0.5	-0.3	-0.6	-0.7	0.1	0.1	1.7	-0.7	-0.7

Item	Label	Marginal χ^2	11	12	13	14
11	I11	0.0				
12	I12	0.0	-0.6			
13	I13	0.0	-0.2	0.5		
14	I14	0.0	0.9	0.3	-0.7	
15	I15	0.0	-0.6	0.0	-0.2	-0.3

E. The likelihood-based values and GOF statistics show the AIC, BIC, M_2 , and RMSEA for the fitted model

Likelihood-based Values and Goodness of Fit Statistics [\(Back to TOC\)](#)

Statistics based on the loglikelihood	
-2loglikelihood:	30501.02
Akaike Information Criterion (AIC):	30625.02
Bayesian Information Criterion (BIC):	30972.28

Statistics based on the full item x item x ... classification

The table is too sparse to compute the general multinomial goodness of fit statistics.

Statistics based on one- and two-way marginal tables

M_2	Degrees of freedom	Probability	RMSEA
153.27	178	0.9101	0.00

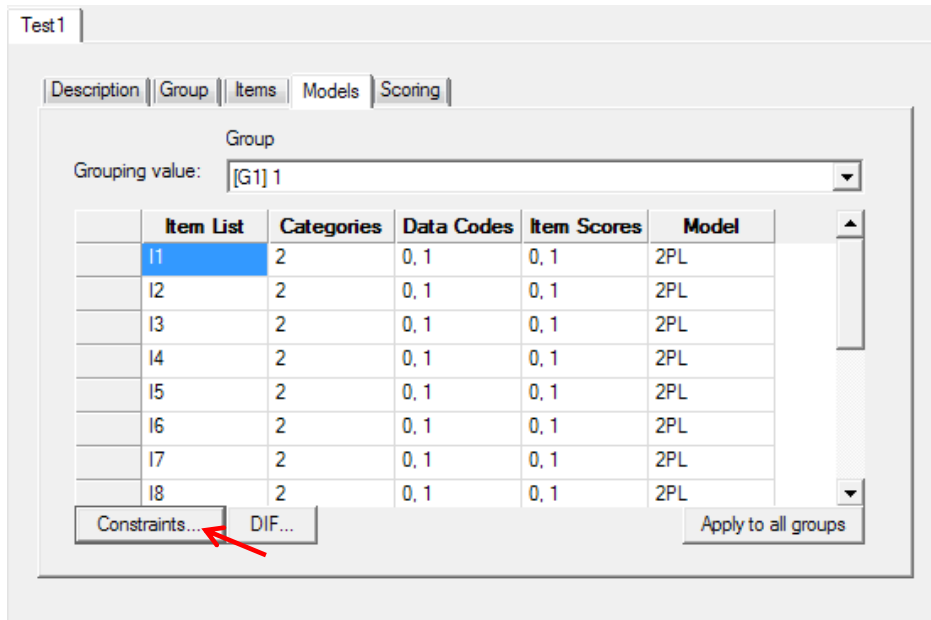
Note: M_2 is based on full marginal tables.

Note: Model-based weight matrix is used.

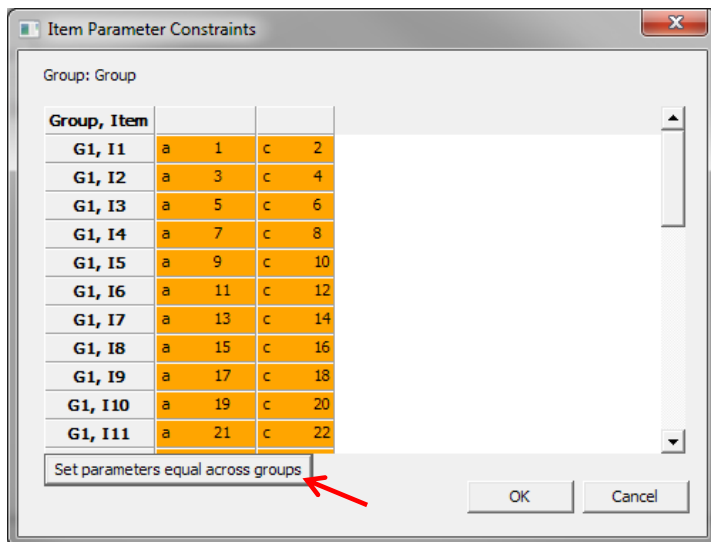
STEP 4: Analyze data using simultaneous estimation (i.e., simultaneous calibration) of both groups: Constraining item parameters to be equal across groups (Model 1)

Follow the same procedure in STEP 2 (A) through (H).

For part (E), click on “Constraints”



Click on “Set parameters equal across groups” > OK



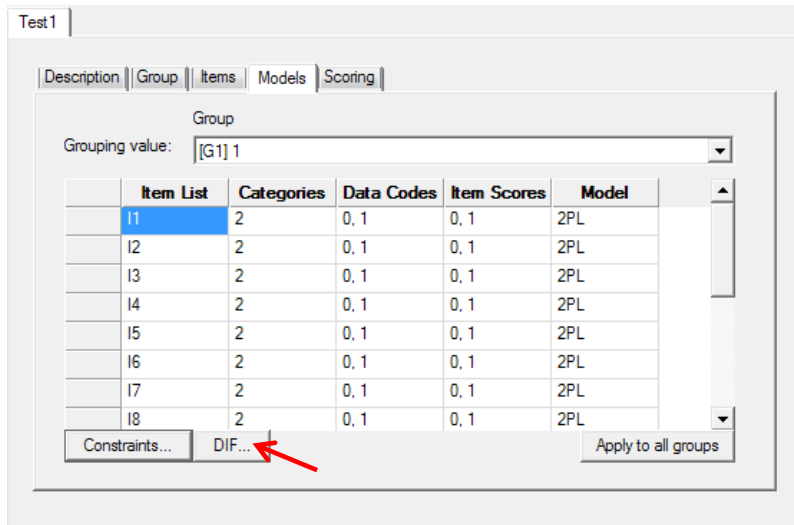
Then “RUN”

This produces a model in which all items are constrained to be equal across groups.

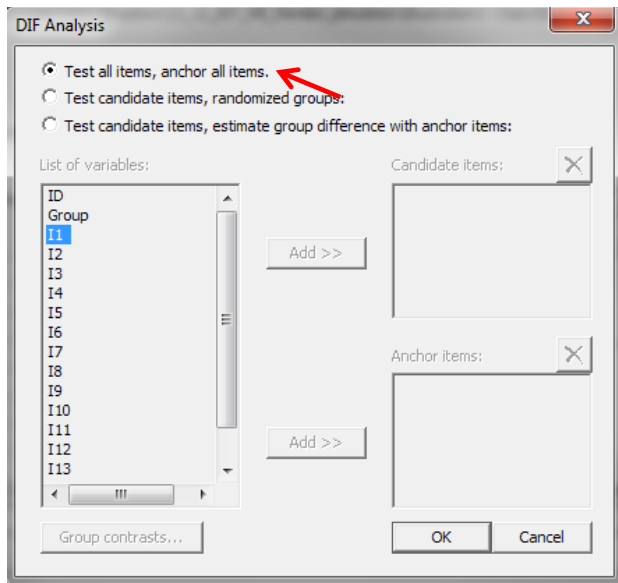
STEP 5: Analyze data using simultaneous estimation (i.e., simultaneous calibration) of both groups: Testing all items for DIF using two-step procedure (Model 2)

Follow the same procedure in STEP 2 (A) through (H).

For part (E), click on “DIF”



Select “Test all items, anchor all items” > OK



Then “OK” > “RUN”

This produces a model in all items are tested for DIF using a two-step procedure. In the first step, all items are assumed to be invariant to estimate focal group latent trait mean and SD. Then, in the next step, all items are freely estimated and focal group latent trait mean and SD are set at the previously estimated values.

In this model, we see that the latent traits are not estimated but fixed. No standard errors are produced for the focal group (Group 2).

Further, we examine the p-values for the Wald χ^2 statistic that tests the difference between reference and focal group item parameters (a* & b). We select items that do not have significant DIF as “anchor items” for our next model (alpha = .05). This includes items 1, 4, 5, 6, 9, 10, 13, & 14.

Group Parameter Estimates [\(Back to TOC\)](#)

Group	Label	μ	s.e.	σ^2	s.e.	σ	s.e.
1	G1	0.00	----	1.00	----	1.00	----
2	G2	-0.08	----	0.50	----	0.71	----

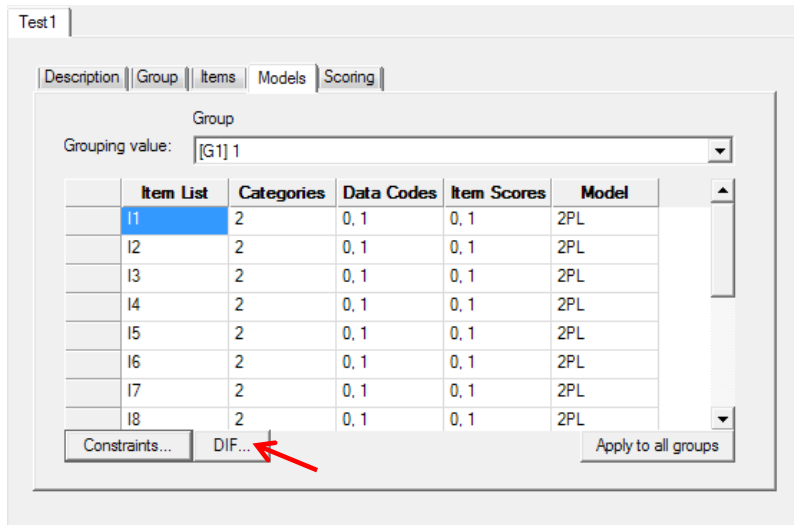
DIF Statistics for Graded Items [\(Back to TOC\)](#)

Item numbers in:											
Group 1	Group 2	Total χ^2	d.f.	p	χ^2_a	d.f.	p	$\chi^2_{c a}$	d.f.	p	
1	1	3.5	2	0.1744	3.4	1	0.0636	0.1	1	0.7966	
2	2	9.7	2	0.0077	9.2	1	0.0025	0.6	1	0.4542	
3	3	55.6	2	0.0001	34.0	1	0.0001	21.6	1	0.0001	
4	4	5.5	2	0.0640	5.1	1	0.0233	0.4	1	0.5535	
5	5	2.1	2	0.3557	2.0	1	0.1575	0.1	1	0.7980	
6	6	1.0	2	0.6099	0.0	1	0.9014	1.0	1	0.3241	
7	7	70.4	2	0.0001	30.2	1	0.0001	40.2	1	0.0001	
8	8	8.9	2	0.0118	7.1	1	0.0077	1.8	1	0.1841	
9	9	4.9	2	0.0879	2.0	1	0.1615	2.9	1	0.0900	
10	10	5.0	2	0.0803	1.7	1	0.1940	3.3	1	0.0672	
11	11	52.8	2	0.0001	31.7	1	0.0001	21.1	1	0.0001	
12	12	7.5	2	0.0231	3.4	1	0.0644	4.1	1	0.0425	
13	13	2.9	2	0.2367	0.6	1	0.4442	2.3	1	0.1300	
14	14	1.8	2	0.4073	1.8	1	0.1861	0.0	1	0.8324	
15	15	48.9	2	0.0001	29.7	1	0.0001	19.2	1	0.0001	

STEP 6: Analyze data using simultaneous estimation (i.e., simultaneous calibration) of both groups: Using anchor items found in Model 2 (Model 3)

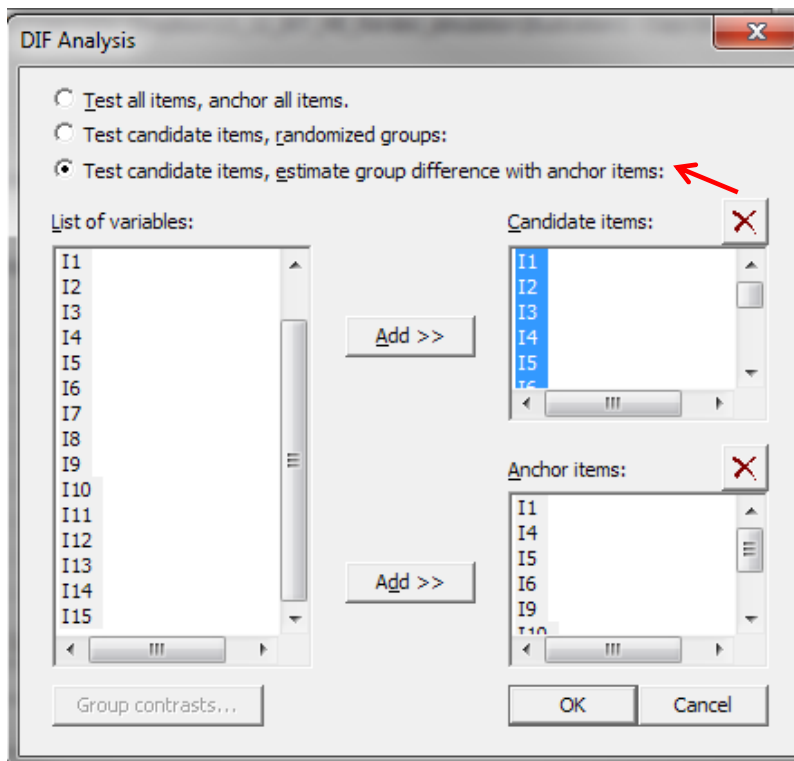
Follow the same procedure in STEP 2 (A) through (H).

For part (E), click on “DIF”



Select “Test candidate items, estimate group difference with anchor items”

Drag all anchor items to the “Anchor items:” box. And all items into “Candidate items:” box.



Then “OK” > “RUN”

This tests for a model in which non-anchor items are tested for DIF.

As shown below, we find the focal group trait mean and SD estimated using the anchor items.

Further, the DIF statistics show that there are a number of non-anchor items that do not have significant DIF (alpha = .05). These include items 2, 8, & 12. We add these as our anchor items at the next step.

Group Parameter Estimates ([Back to TOC](#))

Group	Label	μ	s.e.	σ^2	s.e.	σ	s.e.
1	G1	0.00	-----	1.00	-----	1.00	-----
2	G2	⁴⁵ 0.13	0.05	⁴⁶ 0.89	0.10	⁴⁶ 0.94	0.05

DIF Statistics for Graded Items ([Back to TOC](#))

Item numbers in:											
Group 1	Group 2	Total χ^2	d.f.	p	χ^2_a	d.f.	p	$\chi^2_{cl/a}$	d.f.	p	
1	1	0.0	2	1.0000	0.0	1	1.0000	0.0	1	1.0000	
2	2	0.4	2	0.8358	0.3	1	0.5578	0.0	1	0.9026	
3	3	118.4	2	0.0001	54.5	1	0.0001	63.9	1	0.0001	
4	4	0.0	2	1.0000	0.0	1	1.0000	0.0	1	1.0000	
5	5	0.0	2	1.0000	0.0	1	1.0000	0.0	1	1.0000	
6	6	0.0	2	1.0000	0.0	1	1.0000	0.0	1	1.0000	
7	7	109.1	2	0.0001	45.5	1	0.0001	63.6	1	0.0001	
8	8	1.3	2	0.5279	1.2	1	0.2718	0.1	1	0.7933	
9	9	0.0	2	1.0000	0.0	1	1.0000	0.0	1	1.0000	
10	10	0.0	2	1.0000	0.0	1	1.0000	0.0	1	1.0000	
11	11	138.9	2	0.0001	54.8	1	0.0001	84.1	1	0.0001	
12	12	2.6	2	0.2719	1.8	1	0.1772	0.8	1	0.3767	
13	13	0.0	2	1.0000	0.0	1	1.0000	0.0	1	1.0000	
14	14	0.0	2	1.0000	0.0	1	1.0000	0.0	1	1.0000	
15	15	129.0	2	0.0001	50.2	1	0.0001	78.8	1	0.0001	

STEP 7: Analyze data using simultaneous estimation (i.e., simultaneous calibration) of both groups: Using anchor items found in Model 3 (Model 4)

Follow the same procedure in STEP 6 (A) through (H).

Select the anchor items: 1,2,4,5,6,8,9,10,12,13,14. Test all the other items for DIF.

As shown in the output below, we find that all the non-anchor items have significant DIF. The iterative procedure ends at this point.

DIF Statistics for Graded Items ([Back to TOC](#))

Item numbers in:										
Group 1	Group 2	Total χ^2	d.f.	p	χ^2_a	d.f.	p	$\chi^2_{cl,a}$	d.f.	p
1	1	0.0	2	1.0000	0.0	1	1.0000	0.0	1	1.0000
2	2	0.0	2	1.0000	0.0	1	1.0000	0.0	1	1.0000
3	3	124.7	2	0.0001	58.1	1	0.0001	66.6	1	0.0001
4	4	0.0	2	1.0000	0.0	1	1.0000	0.0	1	1.0000
5	5	0.0	2	1.0000	0.0	1	1.0000	0.0	1	1.0000
6	6	0.0	2	1.0000	0.0	1	1.0000	0.0	1	1.0000
7	7	111.3	2	0.0001	47.6	1	0.0001	63.7	1	0.0001
8	8	0.0	2	1.0000	0.0	1	1.0000	0.0	1	1.0000
9	9	0.0	2	1.0000	0.0	1	1.0000	0.0	1	1.0000
10	10	0.0	2	1.0000	0.0	1	1.0000	0.0	1	1.0000
11	11	149.8	2	0.0001	57.8	1	0.0001	92.0	1	0.0001
12	12	0.0	2	1.0000	0.0	1	1.0000	0.0	1	1.0000
13	13	0.0	2	1.0000	0.0	1	1.0000	0.0	1	1.0000
14	14	0.0	2	1.0000	0.0	1	1.0000	0.0	1	1.0000
15	15	139.7	2	0.0001	52.4	1	0.0001	87.2	1	0.0001

Illustration 2

File Name	Comment
Simulated_PData.csv	Contains simulated data of 2000 individuals. Group = 1 represents the reference group (N = 1000); Group = 2 represents the focal group (N =1000); I1 to I15 represents items 1 to 15. See the simulated item parameters below (Table 8 in paper).
Simulated_PData.irtpro	IRTPRO syntax file
Simulated_PData.SSIG	IRTPRO data file (converted from the .csv file)
Simulated_PData.Model0-irt	Model 0 Output – Simultaneous estimation (no constraints)
Simulated_PData.Model1-irt	Model 1 Output – Fully constrained model
Simulated_PData.Model2-irt	Model 2 Output – Testing anchor items with two-step procedure
Simulated_PData.Model3-irt	Model 3 Output – Testing non-anchor items for DIF
Simulated_PData.Model4-irt	Model 4 Output – Further testing non-anchor items for DIF
Simulated_PData.Model5-irt	Model 5 Output – Further testing non-anchor items for DIF using different contrasts

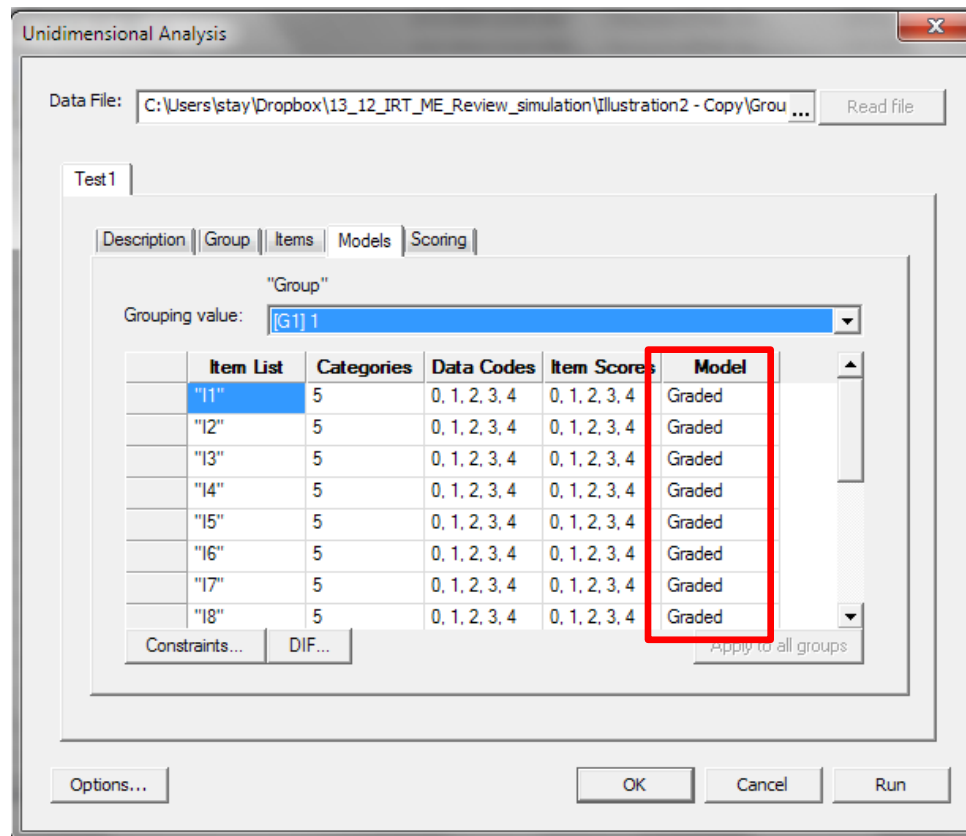
Table 10. Illustration 2: Simulated item and theta parameters

Group 1 ($\theta_{\text{mean}} = 0$; $\theta_{\text{sd}} = 1$)						Group 2 ($\theta_{\text{mean}} = 0$; $\theta_{\text{sd}} = 1$)					Type of DIF	Group 3 ($\theta_{\text{mean}} = -.30$; $\theta_{\text{sd}} = 1$)					Type of DIF
a	b1	b2	b3	b4		a	b1	b2	b3	b4		a	b1	b2	b3	b4	
1	2.06	-1.34	-0.63	-0.29	0.47	2.06	-1.34	-0.63	-0.29	0.47		2.06	-1.34	-0.63	-0.29	0.47	
2	0.88	-2.15	-0.76	-0.09	1.68	0.88	-2.15	-0.76	-0.09	1.68		0.88	-2.15	-0.76	-0.09	1.68	
3	1.49	-2.04	-1.18	-0.51	0.77	0.48	-1.43	-0.58	0.10	1.37	Large ab DIF	0.48	-1.43	-0.58	0.10	1.37	Large ab DIF
4	1.01	-1.80	-0.65	-0.20	0.86	1.01	-1.80	-0.65	-0.20	0.86		0.33	-1.80	-0.65	-0.20	0.86	Large a DIF
5	1.21	-2.03	-1.06	-0.48	0.70	1.21	-2.03	-1.06	-0.48	0.70		1.21	-1.38	-0.42	0.17	1.35	Large b DIF
6	0.93	-2.53	-1.24	-0.50	1.03	0.93	-2.53	-1.24	-0.50	1.03		0.93	-2.53	-1.24	-0.50	1.03	
7	0.71	-2.98	-1.62	-0.83	0.62	0.18	-2.12	-0.76	0.03	1.48	Large ab DIF	0.18	-2.12	-0.76	0.03	1.48	
8	1.33	-1.48	-0.53	-0.09	0.96	1.33	-1.48	-0.53	-0.09	0.96		1.33	-1.48	-0.53	-0.09	0.96	
9	1.61	-1.85	-0.92	-0.35	0.84	1.61	-1.85	-0.92	-0.35	0.84		1.61	-1.85	-0.92	-0.35	0.84	
10	1.61	-1.81	-1.07	-0.56	0.39	1.61	-1.81	-1.07	-0.56	0.39		1.61	-1.81	-1.07	-0.56	0.39	
11	1.43	-1.85	-0.89	-0.33	0.88	0.90	-1.55	-0.59	-0.02	1.18	Small ab DIF	0.90	-1.55	-0.59	-0.02	1.18	Small ab DIF
12	1.33	-1.89	-0.79	-0.33	0.70	0.66	-1.89	-0.79	-0.33	0.70	Small a DIF	0.66	-1.89	-0.79	-0.33	0.70	Small a DIF
13	1.61	-1.40	-0.45	-0.04	1.01	1.61	-1.11	-0.15	0.26	1.31	Small b DIF	1.61	-1.11	-0.15	0.26	1.31	Small b DIF
14	1.55	-1.61	-0.64	-0.17	0.95	1.55	-1.61	-0.64	-0.17	0.95		1.55	-1.61	-0.64	-0.17	0.95	
15	1.69	-1.53	-0.72	-0.31	0.56	0.90	-1.24	-0.43	-0.02	0.85	Small ab DIF	0.90	-1.24	-0.43	-0.02	0.85	Small ab DIF

The same steps shown for Illustration 1 are used.

The three main differences are:

- (i) In STEP 2 (E), for the “Models” tab, the graded response model (GRM) is selected (by default) instead of the 2PLM as the responses are polytomous



- (ii) The testing of DIF in subsequent steps requires the use of contrasts as there are multiple groups. The default two contrasts are

Contrast	Group 1 (Reference Group)	Group 2 (Focal Group 1)	Group 3 (Focal Group 2)	Comment
1	2	-1	-1	Tests whether item parameters in Group 1 differ from Group 2 and 3
2	0	1	-1	Tests whether item parameters in Group 2 differ from Group 3

DIF Statistics for Graded Items [\(Back to TOC\)](#)

Item numbers in:			Contrast	Total χ^2	d.f.	p	χ^2_a	d.f.	p	χ^2_{cla}	d.f.	p
Group 1	Group 2	Group 3										
1	1	1	1	7.3	5	0.1974	0.3	1	0.5915	7.0	4	0.1339
			2	4.8	5	0.4432	3.5	1	0.0613	1.3	4	0.8632
2	2	2	1	10.2	5	0.0691	1.2	1	0.2745	9.0	4	0.0604
			2	1.3	5	0.9329	0.6	1	0.4298	0.7	4	0.9519
3	3	3	1	160.9	5	0.0001	66.4	1	0.0001	94.5	4	0.0001
			2	2.7	5	0.7506	0.0	1	0.8616	2.6	4	0.6199
4	4	4	1	26.3	5	0.0001	5.5	1	0.0191	20.8	4	0.0003
			2	161.1	5	0.0001	64.7	1	0.0001	96.4	4	0.0001
5	5	5	1	10.0	5	0.0747	0.7	1	0.4058	9.3	4	0.0535
			2	63.7	5	0.0001	1.3	1	0.2467	62.3	4	0.0001
6	6	6	1	7.3	5	0.2002	0.0	1	0.9363	7.3	4	0.1219
			2	13.6	5	0.0183	0.0	1	0.8445	13.6	4	0.0088
7	7	7	1	196.3	5	0.0001	26.4	1	0.0001	169.9	4	0.0001
			2	0.9	5	0.9707	0.1	1	0.7771	0.8	4	0.9367
8	8	8	1	5.5	5	0.3628	0.5	1	0.4916	5.0	4	0.2892
			2	3.6	5	0.6163	1.5	1	0.2246	2.1	4	0.7231

The DIF output shows the Wald χ^2 statistic and the associated p-value for the two contrasts. When selecting anchor items, we want to select items that do not show significant p-values for both contrasts. In this sample of 8 items, we see that items 1, 2, and 8 have non-significant p-values across both contrasts.

- (iii) Another difference is that we also specify DIF contrasts apart from using the default values. In the “Models” Tab > “DIF...” > “Group contrasts ...”

	Label	Contrast 1	Contrast 2
1	G1	2	0
2	G2	-1	1
3	G3	-1	-1

☒ Orthogonalize contrasts in the metric of the group sizes
☐ Do not orthogonalize contrasts

OK Cancel

In our illustration, we used the default two contrasts and then used contrasts 3 and 4 to test whether item parameters differ between reference and specific focal groups.

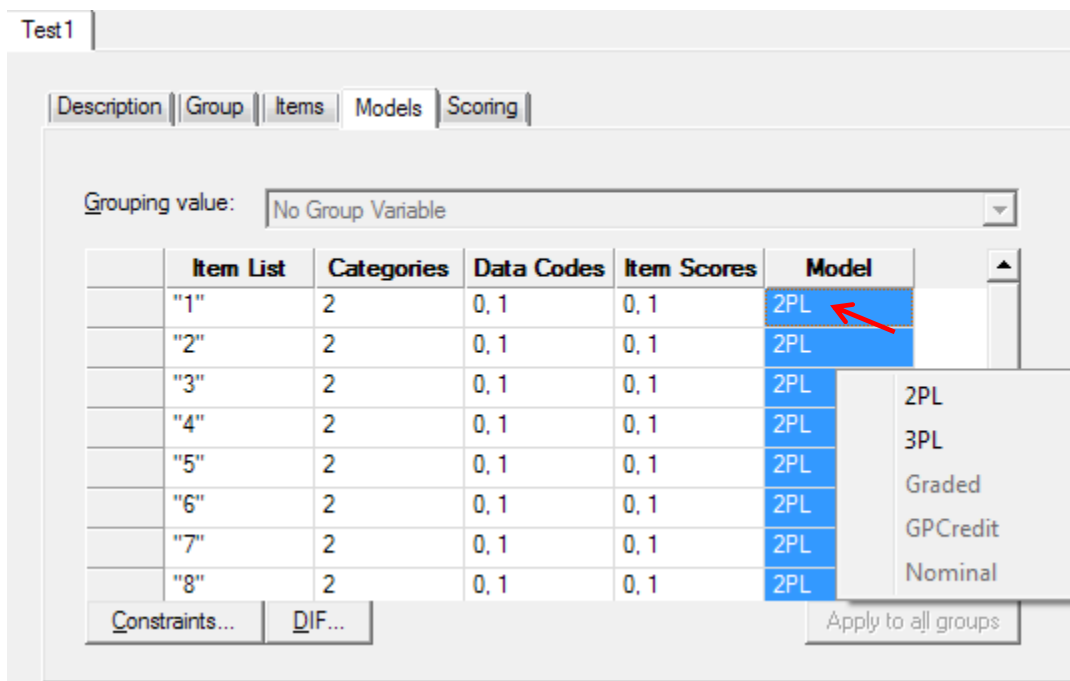
Contrast	Group 1 (Reference Group)	Group 2 (Focal Group 1)	Group 3 (Focal Group 2)	Comment
3	1	-1	0	Tests whether item parameters in Group 1 differ from Group 2
4	1	0	-1	Tests whether item parameters in Group 1 differ from Group 3

Illustration 3

File Name	Comment
Data.sav	Contains simulated data of 5000 individuals. X1 is a dichotomous grouping variable (0, 1) (e.g., gender, Black-White, etc.); X2 is a continuous variable (e.g., age, income, etc.).
Data_Restructure.sav	Restructured Data.sav for 3PLM IRT analysis in LG
Data_Restructure.LGS	Latent GOLD syntax
Simulated_3PL-irt.htm	IRTPRO output

Running IRTPRO to examine model-data fit

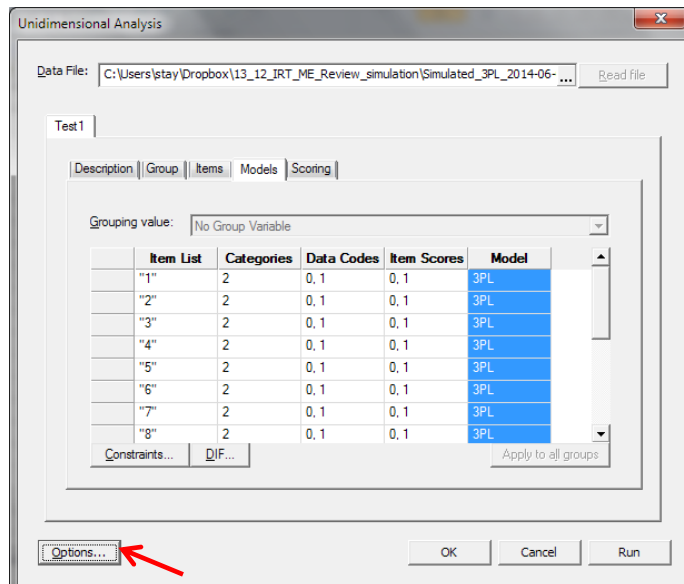
The steps for running IRTPRO to examine model data fit are in line with Illustrations 1 and 2. The difference is that in Illustration 3 we are specifying a 3PLM. As such, in the “Models” tab, we need to change the 2PL to 3PL. This can be done by highlighting all the items and right clicking for additional models. Then we choose 3PL.



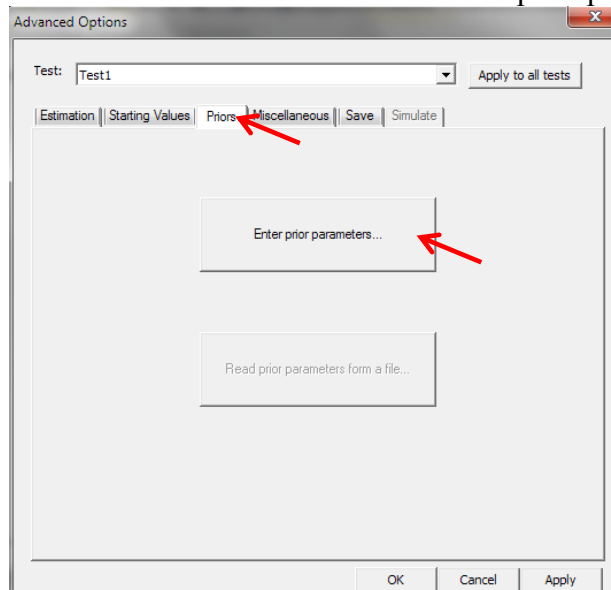
For a 3PLM, it is helpful to specify priors for the c-parameter otherwise it is usually poorly estimated (large standard errors). We can specify a Beta distribution (α , β) for a c-prior. It has been recommended that the values chosen for the Beta distribution are based on these equations: $\alpha = mp + 1$ and $\beta = m(1 - p) + 1$ (Harwell & Baker, 1991). The value of m would range from 15 to 20 depending on the confidence one has in the prior information (higher values indicate higher levels of confidence). In BILOG, m is set at 20 by default. This is the value we use as well. The value of

p is $1/\text{Noptions}$, where Noptions denotes the number of response options there are. For example, if there are 5 options on the test, $p = 1/5 = .20$. There is on average a 20% chance of getting an answer correct with random guessing. Therefore, $\alpha = mp + 1 = 5$; $\beta = m(1-p) + 1 = 17$.

To set the priors, go to “Options...”



Then click on the “Priors” tab > “Enter prior parameters”



Highlight the entire third column of “g values”. This represents the “c-parameters” for the 3PLM. Then right click to choose the “Beta” distribution

Prior Parameter Values

Group: Single Group

Item				
"7"	a	19	c	20
"8"	a	22	c	23
"9"	a	25	c	26
"10"	a	28	c	29
"11"	a	31	c	32
"12"	a	34		
"13"	a	37		
"14"	a	40		
"15"	a	43		
Means	μ_1	0.0		
Covariances	σ_1	1	1.0	

Normal...
Lognormal...
Beta...

OK Cancel

Enter the desired values for the Beta distribution

Priors Values

a-1: 20

b-1: 17

OK Cancel

Running Latent GOLD for DIF analysis

STEP 1: Understanding the LG parameterization

The parameterization for Latent GOLD is different from the parameterization used to simulate the item parameters. In addition, the simulated latent trait values need to be rescaled.

We simulated item parameters in with a scaling factor of “1.702”

$$c_i + \frac{1 - c_i}{1 + \exp[-1.702a_i(\theta_j - [b_i + d_iX1 + e_iX2])]}$$

In addition, the θ_j value simulated is not standardized.

$$\theta_j = 0.00 - .20(X1) + .30(X2) + e$$

Because in the estimation, the latent trait distribution is fixed at N(0,1), we need to divide θ_j by the SD of θ_j , which in this case, the expected value is .58.

$$\frac{\theta_j}{SD} = 0.00 - \frac{.20}{SD}(X1) + \frac{.30}{SD}(X2) + e$$

$$\Rightarrow \frac{\theta_j}{.58} = 0.00 - \frac{.20}{.58}(X1) + \frac{.30}{.58}(X2) + e$$

In the response equation,

$$c_i + \frac{1 - c_i}{1 + \exp[-1.702a_i(\theta_j - [b_i + d_iX1 + e_iX2])]}$$

$$\Rightarrow c_i + \frac{1 - c_i}{1 + \exp[-\left([1.702a_i \times .58] \frac{\theta_j}{.58} + [-1.702a_i]b_i + [-1.702a_i]d_iX1 + [-1.702a_i]e_iX2\right)]}$$

$$\Rightarrow c_i + \frac{1 - c_i}{1 + \exp[-\left([1.702a_i \times .58] \frac{\theta_j}{.58} + [-1.702a_i]b_i + [-1.702a_i]d_iX1 + [-1.702a_i]e_iX2\right)]}$$

$$\Rightarrow c_i + \frac{1 - c_i}{1 + \exp[-(a_i^*\theta_j + b_i^* + d_i^*X1 + e_i^*X2)]}$$

where $a_i^* = 1.702a_i \times .58$

$$b_i^* = -1.702a_i b_i$$

$$d_i^* = -1.702a_i d_i$$

$$e_i^* = -1.702a_i e_i$$

Item parameter conversions from Simulated Item parameters to LG item parameters

Simulated Item parameters					Reparameterized into LG parameters					
	a	b	c	d	e	a*	b*	c*	d*	e*
1	2.06	-0.29	0.15			2.04	1.02	0.15		
2	0.88	-0.09	0.15			0.87	0.14	0.15		
3	1.49	-0.51	0.15	-0.50	0.20	1.47	1.28	0.15	1.27	-0.51
4	1.01	-0.20	0.15			1.00	0.34	0.15		
5	1.21	-0.48	0.15			1.19	0.99	0.15		
6	0.93	-0.50	0.20			0.92	0.79	0.20		
7	0.71	-0.83	0.20	-0.50	0.20	0.70	1.00	0.20	0.61	-0.24
8	1.33	-0.09	0.20		0.30	1.32	0.20	0.20		-0.68
9	1.61	-0.35	0.20			1.59	0.97	0.20		
10	1.61	-0.56	0.20			1.59	1.55	0.20		
11	1.43	-0.33	0.25	-0.25		1.41	0.80	0.25	0.61	
12	1.33	-0.33	0.25		0.50	1.32	0.74	0.25		-1.13
13	1.61	-0.04	0.25		0.50	1.59	0.10	0.25		-1.37
14	1.55	-0.17	0.25			1.53	0.44	0.25		
15	1.69	-0.31	0.25	-0.25		1.66	0.90	0.25	0.72	

STEP 2: Preparing the data for LG analysis

Because the 3PLM is unique in that it has a “guessing” parameter, we need to structure the data in a unique format so that we can use generalized latent variable modeling. Specifically, we need to have a “long” and “wide” format for this analysis.

Specifically, if we have 4 items Y1 to Y4,

ID	Y1	Y2	Y3	Y4
1	0	0	1	1
2	1	0	1	1

We will need to restructure it to the following...

ID	itemnr	response	Y1	Y2	Y3	Y4
1	1	0	.00			
1	2	0		.00		
1	3	1			1.00	
1	4	1				1.00
2	1	1	1.00			
2	2	0		.00		
2	3	1			1.00	
2	4	1				1.00

The SPSS syntax is as follows:

```
VARSTOCASES
  /ID=case
  /MAKE response FROM y1 y2 y3 y4 y5 y6 y7 y8 y9 y10 y11 y12 y13 y14 y15
  /INDEX=itemnr(15)
  /KEEP=x1 x2 ID
  /NULL=KEEP.

IF itemnr = 1 y1 = response.
IF itemnr = 2 y2 = response.
IF itemnr = 3 y3 = response.
IF itemnr = 4 y4 = response.
IF itemnr = 5 y5 = response.
IF itemnr = 6 y6 = response.
IF itemnr = 7 y7 = response.
IF itemnr = 8 y8 = response.
IF itemnr = 9 y9 = response.
IF itemnr = 10 y10 = response.
IF itemnr = 11 y11 = response.
IF itemnr = 12 y12 = response.
IF itemnr = 13 y13 = response.
IF itemnr = 14 y14 = response.
IF itemnr = 15 y15 = response.
EXECUTE.
```

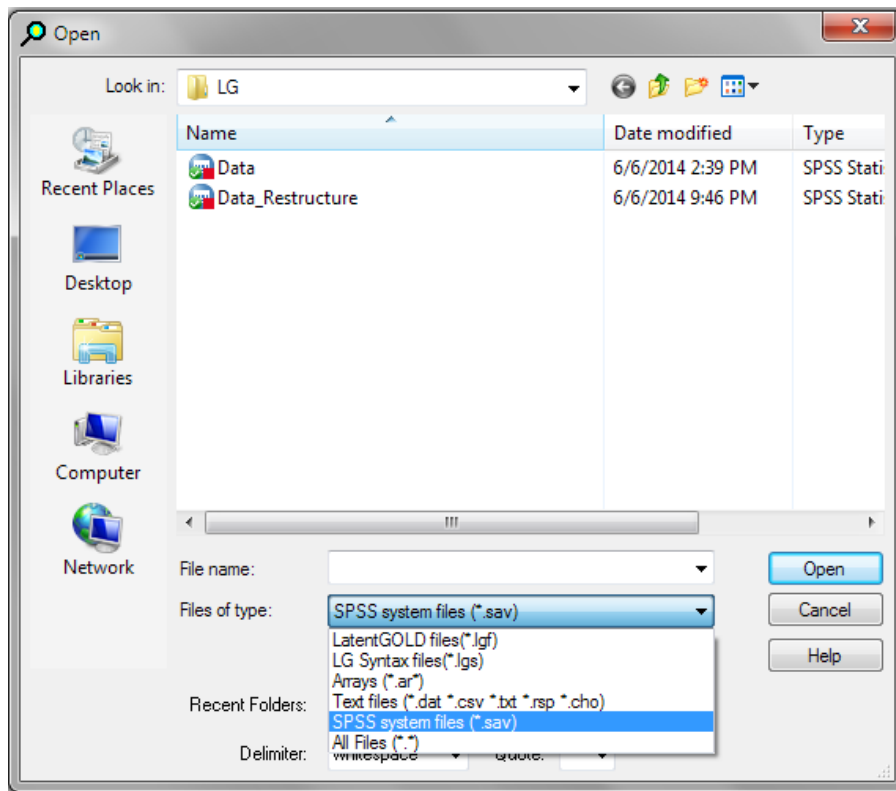
Using this restructured data, we can then proceed to analyze it in Latent GOLD.

For other models without the “guessing” parameter such as the 1PLM, 2PLM, and GRM, we do not need to have this unique format. We will show some example syntax for these other models in the last section.

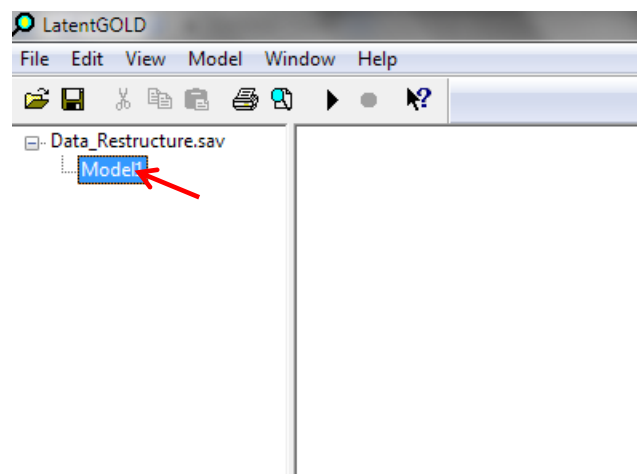
STEP 3: LG 3PL DIF analysis

The proposed procedure is based on research of the IRT-C DIF analysis (Tay, Newman, & Vermunt, 2011; Tay, Vermunt, & Wang, 2013).

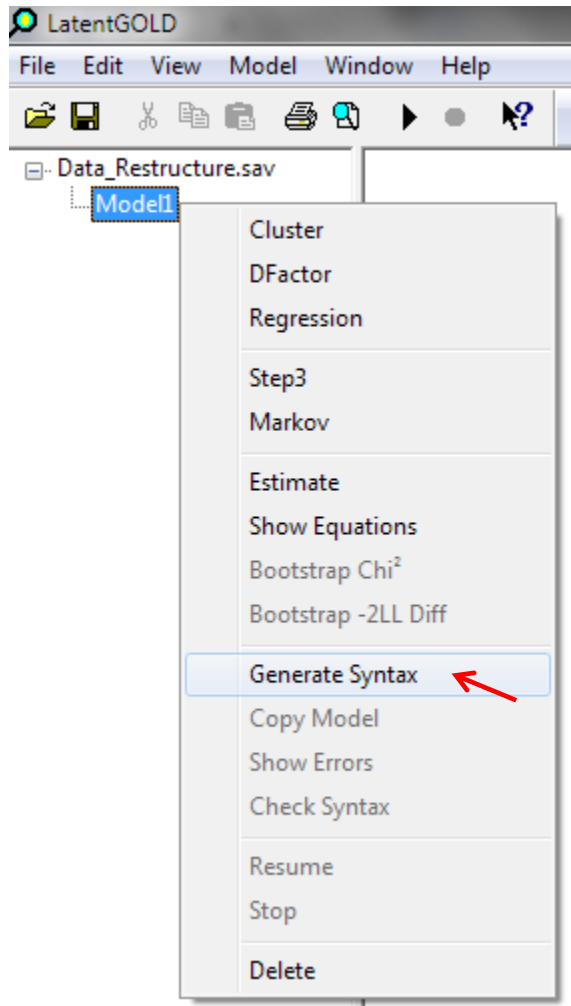
To open the data file in Latent GOLD, we click on “Open”  symbol and select the restructured data file. In this case, we have labeled our restructured data “Data_Restructure.sav”.



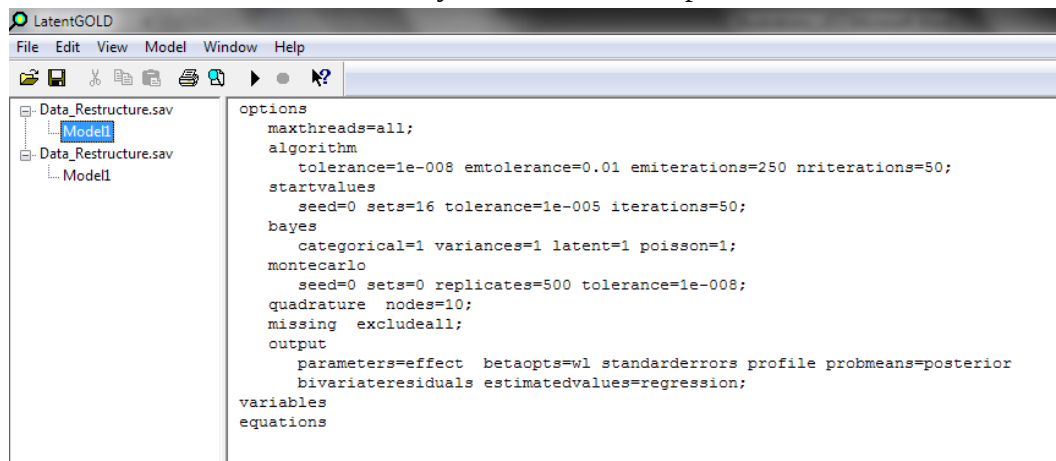
After selecting the file, we should see that it is read in. Then right click on “Model1”



We should see a drop down box after right clicking “Model1”. Select “Generate Syntax” as we want to use the “Syntax” mode.



We should now see that there is syntax in the black space that we can edit.



For a fully constrained model – where all items are constrained as equal across groups...

```
options
  algorithm
    bhhh
    tolerance=1e-008 emtolerance=0.01 emiterations=1000 nriterations=500;
  startvalues
    seed=0 sets=0 tolerance=1e-005 iterations=50;
  bayes
    categorical=1 variances=1 latent=1 poisson=1;
  montecarlo
    seed=0 replicates=500 tolerance=1e-008;
  quadrature nodes=30;
  missing includeall;
  output
    parameters=first standarderrors=fast estimatedvalues bivariateresiduals;
variables
  caseid id;
  dependent y1, y2, y3, y4, y5, y6, y7, y8, y9, y10, y11, y12, y13, y14,
y15;
  independent itemnr nominal, x1, x2 rank=5;
  latent
    theta continuous,
    c dynamic nominal 2;
equations
  (1) theta;
  theta <- x1 + x2 ;
  c <- 1 | itemnr;
  y1 <- 1 + (+) theta + (100) c;
  y2 <- 1 + (+) theta + (100) c;
  y3 <- 1 + (+) theta + (100) c;
  y4 <- 1 + (+) theta + (100) c;
  y5 <- 1 + (+) theta + (100) c;
  y6 <- 1 + (+) theta + (100) c;
  y7 <- 1 + (+) theta + (100) c;
  y8 <- 1 + (+) theta + (100) c;
  y9 <- 1 + (+) theta + (100) c;
  y10 <- 1 + (+) theta + (100) c;
  y11 <- 1 + (+) theta + (100) c;
  y12 <- 1 + (+) theta + (100) c;
  y13 <- 1 + (+) theta + (100) c;
  y14 <- 1 + (+) theta + (100) c;
  y15 <- 1 + (+) theta + (100) c;
```

In the “parameter” tab, the estimated regression weights for the group characteristics (x1 and x2) and the item parameters a* values and b* values are displayed...

Here, we see that being in X1 (0 = reference; 1 = focal) is associated with -.26 lower latent trait.

Here, we see that have a higher value in X2 (which is standardized) for 1SD is associated with .49 higher latent trait.

For the first item, the b* value is .85; the a* value is 1.94

LatentGOLD

File Edit View Model Window Help

Data_Restructure.sav

- Model1 - L² = 28056.3132
 - Syntax
 - Parameters
 - Bivariate Residuals
 - EstimatedValues-Model
- Model2 - L² = 27671.2732
 - Syntax
 - Parameters
 - Bivariate Residuals
 - EstimatedValues-Model
- Model3

Data_Restructure.sav

- Model1

Data_Restructure.sav

- Model1

term		coef	s.e.	z-value	p-value	Wald(0)	df	p-value	Wald(+)	df	p-value
theta	← x1	-0.2624	0.0358	-7.3292	2.3e-13	53.7168	1	2.3e-13			
theta	← x2	0.4949	0.0208	23.8250	1.8e-125	567.6290	1	1.8e-125			
c(1)	← 1	itemnr(1)	0.0000	.	.	116.7188	15	8.2e-18	6.1073	14	0.96
c(2)	← 1	itemnr(1)	-1.5336	0.2920	-5.2528	1.5e-7					
c(1)	← 1	itemnr(2)	0.0000	.	.						
c(2)	← 1	itemnr(2)	-5.1245	25.4211	-0.2016	0.84					
c(1)	← 1	itemnr(3)	0.0000	.	.						
c(2)	← 1	itemnr(3)	-1.0226	1.2346	-0.8283	0.41					
c(1)	← 1	itemnr(4)	0.0000	.	.						
c(2)	← 1	itemnr(4)	-1.1988	0.4294	-2.7920	0.0053					
c(1)	← 1	itemnr(5)	0.0000	.	.						
c(2)	← 1	itemnr(5)	-5.4117	30.1530	-0.1795	0.86					
c(1)	← 1	itemnr(6)	0.0000	.	.						
c(2)	← 1	itemnr(6)	-3.4611	7.8990	-0.4382	0.66					
c(1)	← 1	itemnr(7)	0.0000	.	.						
c(2)	← 1	itemnr(7)	-3.7533	23.5490	-0.1594	0.87					
c(1)	← 1	itemnr(8)	0.0000	.	.						
c(2)	← 1	itemnr(8)	-0.7776	0.2458	-3.1628	0.0016					
c(1)	← 1	itemnr(9)	0.0000	.	.						
c(2)	← 1	itemnr(9)	-1.6269	0.5334	-3.0499	0.0023					
c(1)	← 1	itemnr(10)	0.0000	.	.						
c(2)	← 1	itemnr(10)	-1.3569	0.5525	-2.4562	0.014					
c(1)	← 1	itemnr(11)	0.0000	.	.						
c(2)	← 1	itemnr(11)	-0.9593	0.4346	-2.2072	0.027					
c(1)	← 1	itemnr(12)	0.0000	.	.						
c(2)	← 1	itemnr(12)	-1.0321	1.4897	-0.6928	0.49					
c(1)	← 1	itemnr(13)	0.0000	.	.						
c(2)	← 1	itemnr(13)	-1.7291	2.5091	-0.6891	0.49					
c(1)	← 1	itemnr(14)	0.0000	.	.						
c(2)	← 1	itemnr(14)	-0.9790	0.2264	-4.3250	1.5e-5					
c(1)	← 1	itemnr(15)	0.0000	.	.						
c(2)	← 1	itemnr(15)	-1.0698	.	.						
y1(0)	← 1		0.0000	.	.	52.0536	1	5.4e-13			
y1(1)	← 1		0.8457	0.1172	7.2148	5.4e-13					
y1	← theta		1.9381	0.2003	9.6768	3.8e-22	93.6395	1	3.8e-22		

b* for item 1

a* for item 1

LatentGOLD

File Edit View Model Window Help

Data_Restructure.sav

- Model1 - $L^2 = 28056.3132$
 - Syntax
 - Parameters
 - Bivariate Residuals
 - EstimatedValues-Model
- Model2 - $L^2 = 27671.2732$
 - Syntax
 - Parameters
 - Bivariate Residuals
 - EstimatedValues-Model
- Model3

Data_Restructure.sav

- Model1

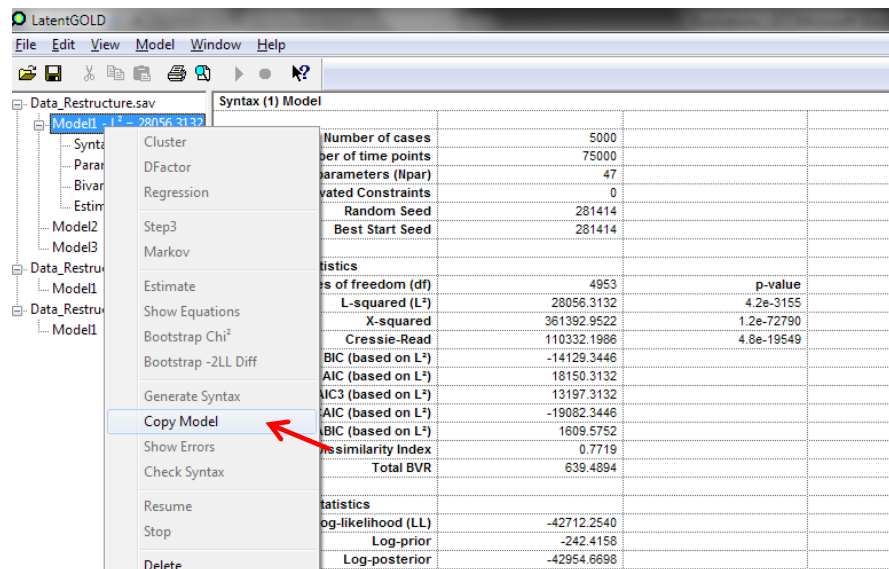
Data_Restructure.sav

Model1

	itemnr	c	1	2
	1	0.8225	0.1775	
	2	0.9941	0.0059	
	3	0.7355	0.2645	
	4	0.7683	0.2317	
	5	0.9956	0.0044	
	6	0.9696	0.0304	
	7	0.9771	0.0229	
	8	0.6852	0.3148	
	9	0.8357	0.1643	
	10	0.7953	0.2047	
	11	0.7230	0.2770	
	12	0.7373	0.2627	
	13	0.8493	0.1507	
	14	0.7269	0.2731	
	15	0.7446	0.2554	

[illegible]

We then proceed to create a model where we allow for DIF for Item13 on covariate X2. This is done by right clicking “Model 1” followed by “Copy Model” which automatically generates “Model 2” with the same exact syntax as “Model 1” for editing.



We edit the equations for Model 2 by adding X2 to equation y13. This effective models for uniform DIF of item 13 on X2. This revised equation shows that responses on y13 are not merely a function of the underlying “theta” trait value, but also dependent on group characteristic X2 (e.g., income, GPA, socioeconomic status, etc.).

```

equations
(1) theta;
theta <- x1 + x2 ;
c <- 1 | itemnr;
y1 <- 1 + (+) theta + (100) c;
y2 <- 1 + (+) theta + (100) c;
y3 <- 1 + (+) theta + (100) c;
y4 <- 1 + (+) theta + (100) c;
y5 <- 1 + (+) theta + (100) c;
y6 <- 1 + (+) theta + (100) c;
y7 <- 1 + (+) theta + (100) c;
y8 <- 1 + (+) theta + (100) c;
y9 <- 1 + (+) theta + (100) c;
y10 <- 1 + (+) theta + (100) c;
y11 <- 1 + (+) theta + (100) c;
y12 <- 1 + (+) theta + (100) c;
y13 <- 1 + (+) theta + x2 + (100) c;
y14 <- 1 + (+) theta + (100) c;
y15 <- 1 + (+) theta + (100) c;

```

We run this model to examine whether the parameter associated with X2 on the equation `y13 <- 1 + (+) theta + x2 + (100) c;` is significant, demonstrating the uniform DIF is significant.

In the “Parameters” tab, we can scroll down to see that -1.0771 is the DIF parameter for X2 and it is significantly different from zero at 3.7e-16.

LatentGOLD

File Edit View Model Window Help

Save Undo Copy Paste Print Help

Data_Restructure.sav

Model1 - L² = 28056.3132

Model2 - L² = 27671.2732

- Syntax
- Parameters
- Bivariate Residuals
- EstimatedValues-Model

Model3

Data_Restructure.sav

Model1

Data_Restructure.sav

Model1

y10	←	c(2)		100.0000	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
y11(0)	←	1		0.0000	.	.	.	.	22.3655	1	2.3e-6	.	.	.	.	.	.	.	.
y11(1)	←	1		0.9766	0.2065	4.7292	2.3e-6	.	.	.	.	.	.	.	.	.	.	.	.
y11	←	theta		1.1793	0.1510	7.8082	5.8e-15	60.9676	1	5.8e-15	.	.	.	.	.	.	.	.	.
y11	←	c(1)		0.0000	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
y11	←	c(2)		100.0000	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
y12(0)	←	1		0.0000	.	.	.	.	0.5212	1	0.47	.	.	.	.	.	.	.	.
y12(1)	←	1		0.4936	0.6837	0.7220	0.47	.	.	.	.	.	.	.	.	.	.	.	.
y12	←	theta		0.4969	0.1567	3.1703	0.0015	10.0511	1	0.0015	.	.	.	.	.	.	.	.	.
y12	←	c(1)		0.0000	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
y12	←	c(2)		100.0000	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
y13(0)	←	1		0.0000	.	.	.	.	0.8227	1	0.36	.	.	.	.	.	.	.	.
y13(1)	←	1		0.1659	0.1829	0.9071	0.36	.	.	.	.	.	.	.	.	.	.	.	.
y13	←	theta		1.1407	0.1438	7.9334	2.1e-15	62.9391	1	2.1e-15	.	.	.	.	.	.	.	.	.
y13	←	x2		-1.0771	0.1322	-8.1486	3.7e-16	66.3997	1	3.7e-16	.	.	.	.	.	.	.	.	.
y13	←	c(1)		0.0000	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
y13	←	c(2)		100.0000	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.

Because it is significant, we then proceed to examine the BVRs again to look for the next item-covariate pair that has the largest BVR.

We continue testing for DIF in this manner until we find that the highest flagged BVR value is no longer significant.

Syntax for 1PLM (No DIF)

```
equations
  (1) theta;
  theta <- x1 + x2 ;
  y1 <- 1 + (1) theta;
  y2 <- 1 + (1) theta;
  y3 <- 1 + (1) theta;
  y4 <- 1 + (1) theta;
  y5 <- 1 + (1) theta;
  y6 <- 1 + (1) theta;
  y7 <- 1 + (1) theta;
  y8 <- 1 + (1) theta;
  y9 <- 1 + (1) theta;
  y10 <- 1 + (1) theta;
  y11 <- 1 + (1) theta;
  y12 <- 1 + (1) theta;
  y13 <- 1 + (1) theta;
  y14 <- 1 + (1) theta;
  y15 <- 1 + (1) theta;
```

Syntax for 1PLM (DIF on Item 13 for covariate x1)

```
equations
  (1) theta;
  theta <- x1 + x2 ;
  y1 <- 1 + (1) theta;
  y2 <- 1 + (1) theta;
  y3 <- 1 + (1) theta;
  y4 <- 1 + (1) theta;
  y5 <- 1 + (1) theta;
  y6 <- 1 + (1) theta;
  y7 <- 1 + (1) theta;
  y8 <- 1 + (1) theta;
  y9 <- 1 + (1) theta;
  y10 <- 1 + (1) theta;
  y11 <- 1 + (1) theta;
  y12 <- 1 + (1) theta;
  y13 <- 1 + (1) theta + x1;
  y14 <- 1 + (1) theta;
  y15 <- 1 + (1) theta;
```

Syntax for 2PLM (No DIF)

```
equations
  (1) theta;
  theta <- x1 + x2 ;
  y1 <- 1 + (+) theta;
  y2 <- 1 + (+) theta;
  y3 <- 1 + (+) theta;
  y4 <- 1 + (+) theta;
  y5 <- 1 + (+) theta;
  y6 <- 1 + (+) theta;
  y7 <- 1 + (+) theta;
  y8 <- 1 + (+) theta;
  y9 <- 1 + (+) theta;
  y10 <- 1 + (+) theta;
  y11 <- 1 + (+) theta;
  y12 <- 1 + (+) theta;
  y13 <- 1 + (+) theta;
  y14 <- 1 + (+) theta;
  y15 <- 1 + (+) theta;
```

Syntax for 2PLM (DIF on Item 13 for covariate x1)

```
equations
  (1) theta;
  theta <- x1 + x2 ;
  y1 <- 1 + (+) theta;
  y2 <- 1 + (+) theta;
  y3 <- 1 + (+) theta;
  y4 <- 1 + (+) theta;
  y5 <- 1 + (+) theta;
  y6 <- 1 + (+) theta;
  y7 <- 1 + (+) theta;
  y8 <- 1 + (+) theta;
  y9 <- 1 + (+) theta;
  y10 <- 1 + (+) theta;
  y11 <- 1 + (+) theta;
  y12 <- 1 + (+) theta;
  y13 <- 1 + (+) theta + x1;
  y14 <- 1 + (+) theta;
  y15 <- 1 + (+) theta;
```

Note: the Graded response model equations are the same for the 2PLM. The only difference is that the responses are “cumlogit”.

```
variables
  dependent y1 cumlogit, y2 cumlogit, y3 cumlogit, y4 cumlogit,
y5 cumlogit, y6 cumlogit, y7 cumlogit, y8 cumlogit, y9 cumlogit,
y10 cumlogit, y11 cumlogit, y12 cumlogit, y13 cumlogit, y14
cumlogit, y15 cumlogit;
  independent itemnr nominal, x1, x2 rank=5;
  latent
    theta continuous;
equations
  (1) theta;
  theta <- x1 + x2 ;
  y1 <- 1 + (+) theta;
  y2 <- 1 + (+) theta;
  y3 <- 1 + (+) theta;
  y4 <- 1 + (+) theta;
  y5 <- 1 + (+) theta;
  y6 <- 1 + (+) theta;
  y7 <- 1 + (+) theta;
  y8 <- 1 + (+) theta;
  y9 <- 1 + (+) theta;
  y10 <- 1 + (+) theta;
  y11 <- 1 + (+) theta;
  y12 <- 1 + (+) theta;
```

Reference

- Harwell, M. R., & Baker, F. B. (1991). The use of prior distributions in marginalized Bayesian item parameter estimation: A didactic. *Applied Psychological Measurement, 15*, 375-389.
- Tay, L., Newman, D. A., & Vermunt, J. K. (2011). Using mixed-measurement item response theory with covariates (MM-IRT-C) to ascertain observed and unobserved measurement equivalence. *Organizational Research Methods, 14*, 147-176. doi: 10.1177/1094428110366037
- Tay, L., Vermunt, J. K., & Wang, C. (2013). Assessing the item response theory with covariate (IRT-C) procedure for ascertaining DIF. *International Journal of Testing*. doi: 10.1080/15305058.2012.692415